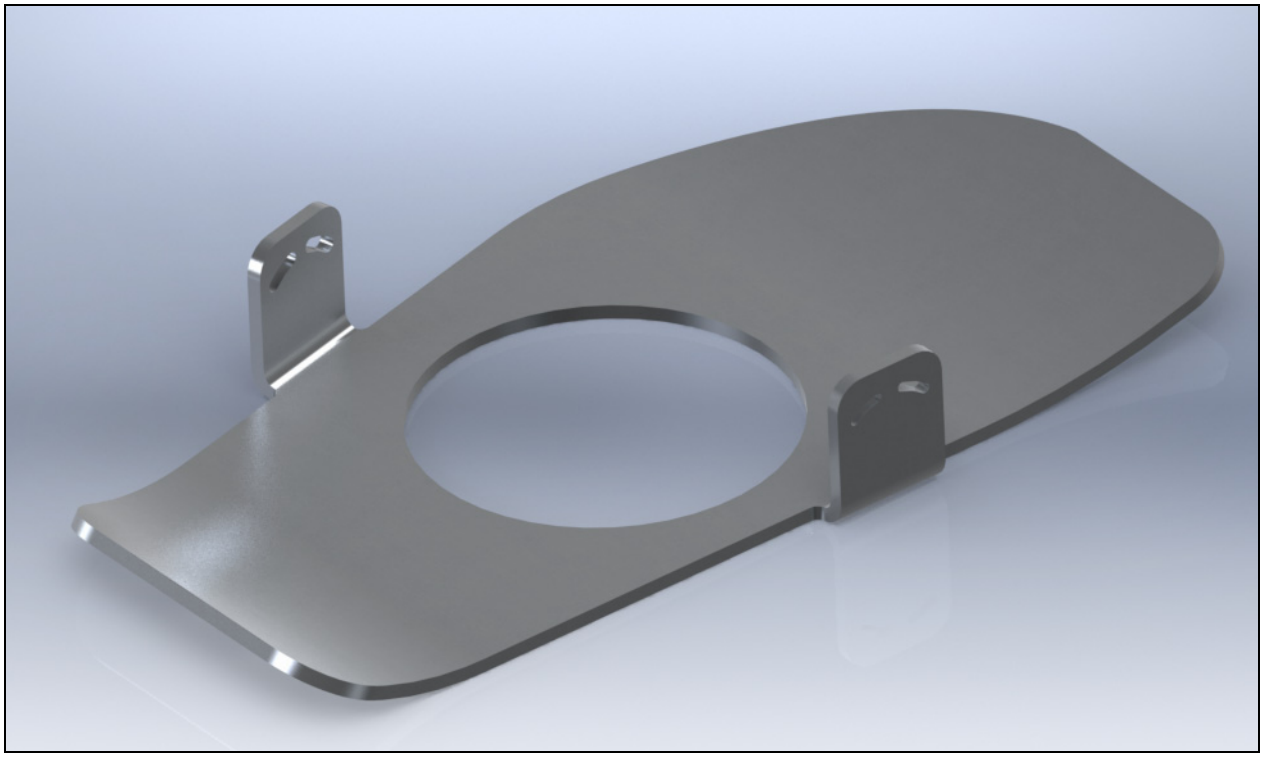
3

Lesson 3: Basic Parts — The Binding

- Students will be able to create and modify the following part:



Before Beginning This Lesson

- Complete the previous lesson: Basic Functionality.

Resources for This Lesson

This lesson plan corresponds to the following lessons in the SolidWorks Tutorial:

- *Lesson 1 – Parts*
- *Sheet Metal*
- *Assembly Mates*

For more information about the Tutorials, See “SolidWorks Tutorials” on page vii.

Review of Lesson 2: Basic Functionality

Questions for Discussion

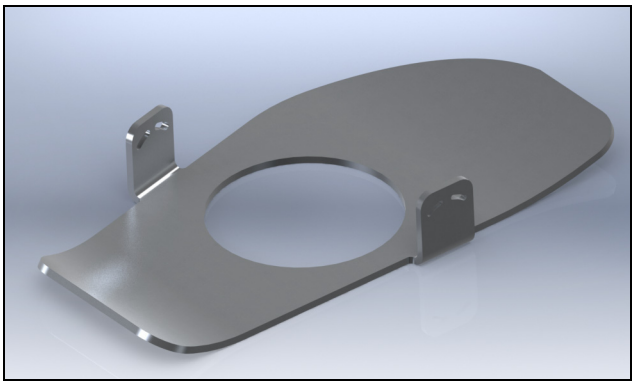
1. A SolidWorks 3D model consists of three documents. Name the three documents.
2. Parts are built from features. What are features?
3. Name the features that are used to create the `Binding Anchor` in Lesson 2.
4. What is the base feature of the `Binding Anchor`?
5. Why did you use the Fillet feature?
6. How did you create the Base feature?

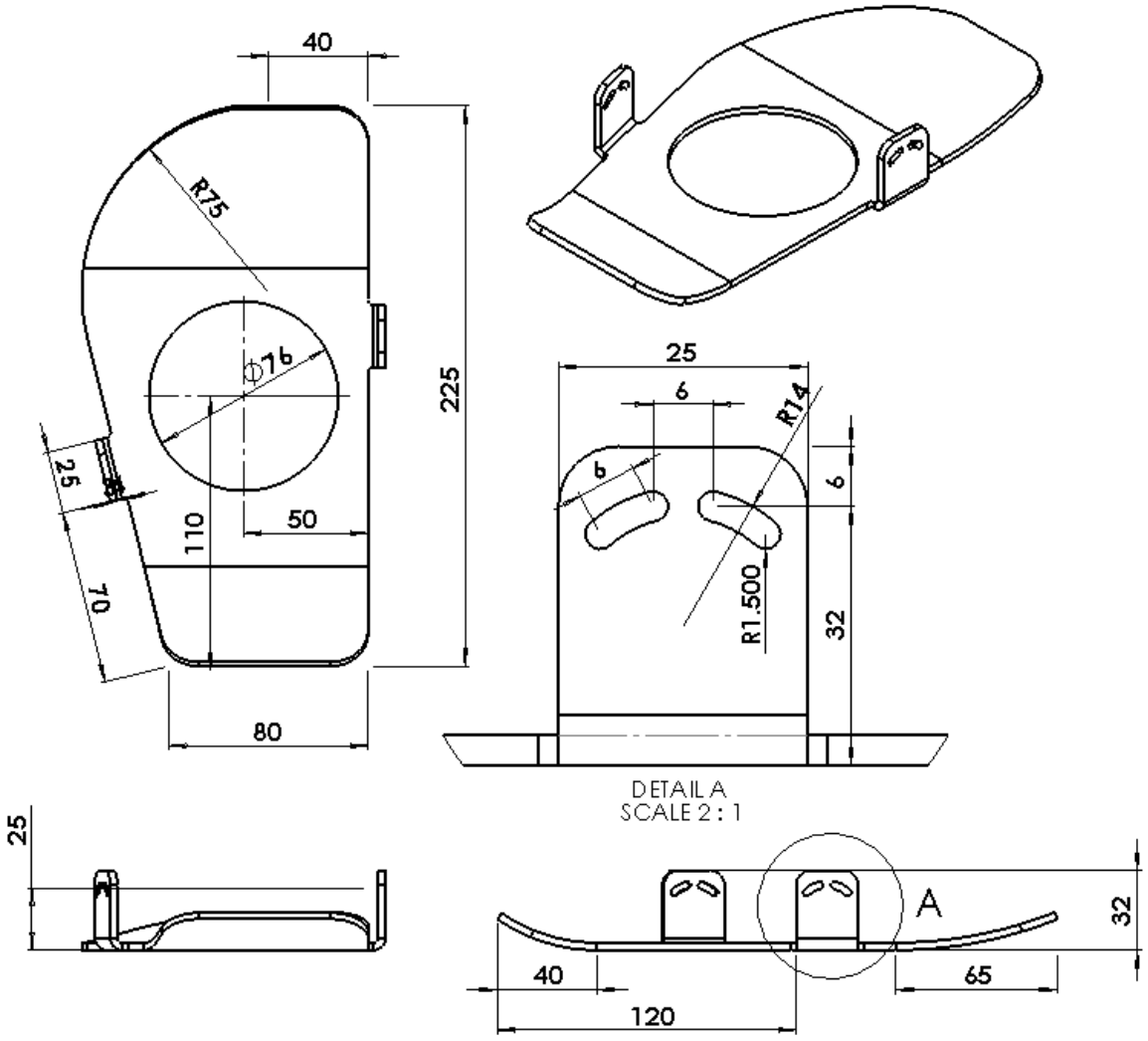
Outline of Lesson 3

- Active Learning Exercise, Part 1— Creating a part
- Active Learning Exercise, Part 2 — Create an assembly
- Exercises and Projects
- Lesson Summary

Active Learning Exercise, Part 1 — Create a Part

Follow the instructions in this lesson to create the `Binding Base Plate`. The `Binding Anchor` created in the last lesson will fit into the large hole in the center to hold the `Binding Base Plate` to the deck of the Mountainboard.





Design Intent

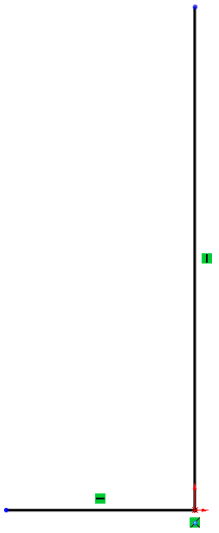
- There will be two versions of this part, one for the left foot and mirror part for the right foot.
- The part will be held in place by the `Binding Anchor` created in the last lesson.
- The front and back of the part will curve upward.
- Side tabs will help hold the foot centered on the binding.
- The side tabs will have slots to attach the flexible straps that go over the foot.

The Modeling Approach

The `Binding Anchor`, created in the previous lesson, was created by adding material on top of material to get the basic geometry. For the `Binding Base Plate`, the approach will be to create an oversized block of material and use a "cookie cutter" to cut away the material around the final part.

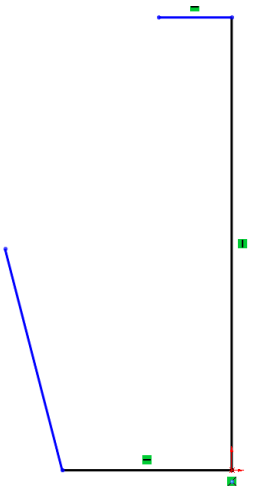
Task 1— Create the First Feature

1. Create a new part using the template `Part_MM.slddot`.
2. Create a sketch on the `Top` plane.
3. Click **View, Sketch Relations** to make the callouts visible.
4. From the origin, create a vertical and horizontal line as shown. The vertical line should be about 200 mm and the horizontal line 75 mm.



5 Create another horizontal line from the end of the vertical line. Make this line about 35mm.

6 Create a line from the end of the first horizontal line. Make sure that this line is *NOT* vertical.

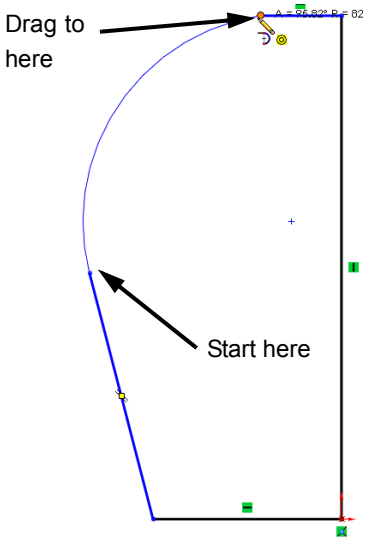


Tangent Arc

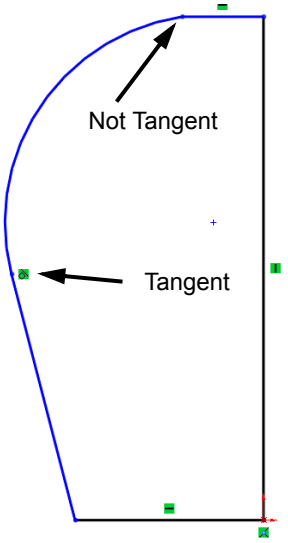
Tangent Arcs are used to create an arc that begins tangent to a selected endpoint on the sketch.

7 Click **Tangent Arc** on the **Sketch** toolbar.

8 Draw an arc from the endpoint of the angled line to the end of the top horizontal line.

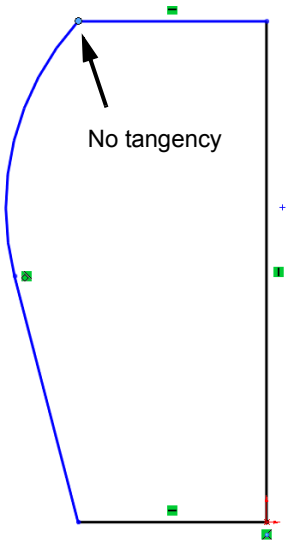


9 Check the relationships of the arc. There was a tangent relationship added between the arc and the line where the arc started.



10 Add a tangent relationship. Notice that there is no tangent relationship between the arc and the top horizontal line. The automatic tangent relationship is only added to the starting end of the arc. To test this, drag the left end of the upper horizontal line to the left.

If a tangent relationship is required at the finish end, it must be added manually.



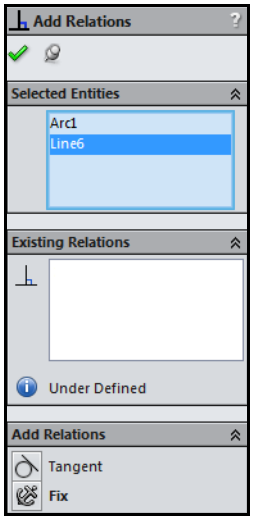
Adding Sketch Relationships

The **Add Relation** tool allows us to add relationships to geometry after it has been created. Only relationships that are appropriate for the selected geometry will be shown.

11 Click on the **Sketch** toolbar.

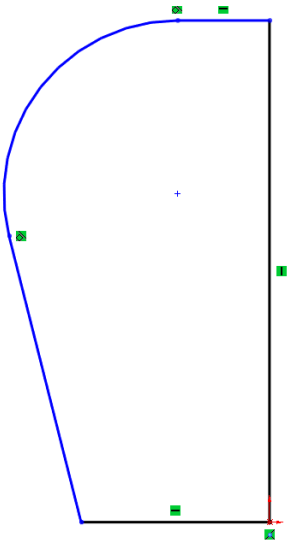
12 Select the arc and the upper horizontal line.

13 The only two choices available are **Tangent** and **Fix** as these are the only two relationships that can be established between an arc and a line.



14 Click **Tangent** . The arc and upper horizontal line are now tangent. Check the relationships by dragging the same point as in the earlier step. The tangency will be maintained.

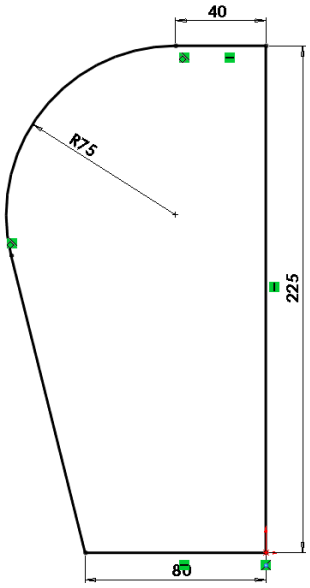
15 The callouts will show that the arc is now tangent at both ends.



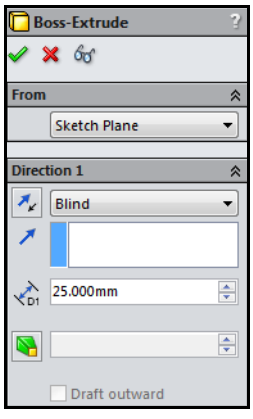
16 Fully define the sketch by adding the dimensions shown.

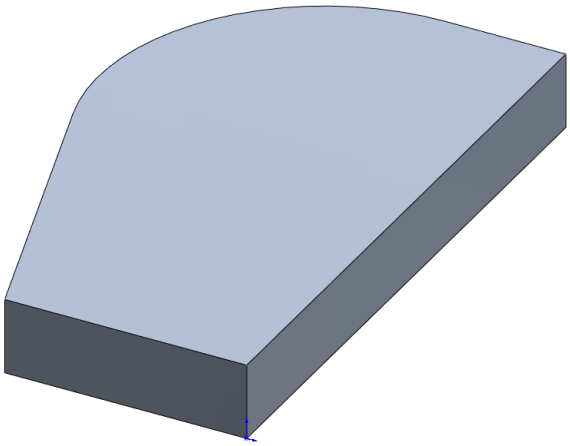
17 All the sketch geometry should now be black.

18 Extrude the feature. Click on the **Features** toolbar.



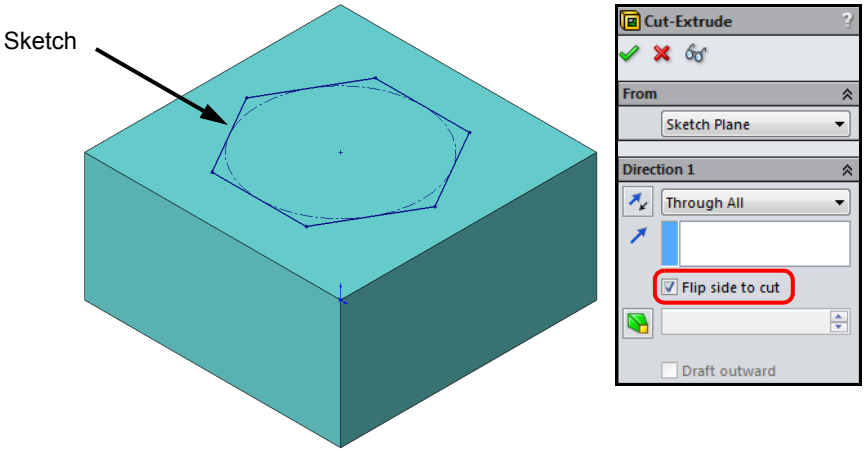
19 Create a **Blind** extrusion to a depth of **25mm**.

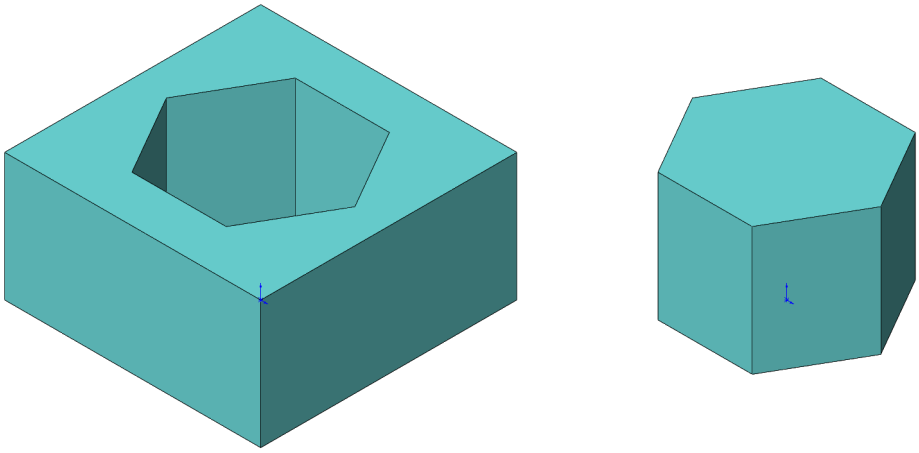




Flip Side To Cut

Normally, we use the **Extruded Cut** to remove the material inside a sketch. It can also be used to remove all the material outside a sketch. This is like using a cookie cutter where we are interested in keeping the shape inside the cutter. Selecting or clearing **Flip Side To Cut** determines if the material inside or outside of the sketch is removed.





Flip Side To Cut - Cleared

Flip Side To Cut - Selected

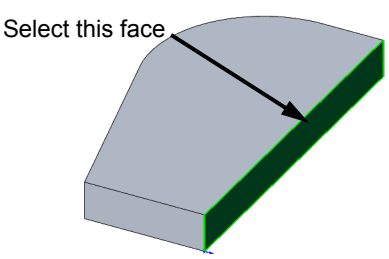
Task 2 — Cookie Cutter Cuts

To get the shape we are interested in, we will use an **Extruded Cut** that will remove the unwanted material from our base feature.

Changing View Orientation

Is is frequently easier to sketch when looking directly at the sketch plane. To change the view orientation to look directly at the sketch plane, select the sketch plane then click **Normal to**  on the **Standard Views** toolbar.

Create a Sketch



1 Select the face shown and open a new sketch.

2 Change the view so that you are looking normal to the sketch plane. Click on the **Standard** Views toolbar.

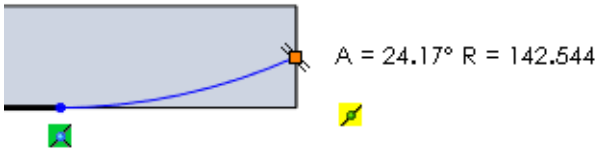
3 Select the **Line** tool.

4 Position the cursor over the lower edge of the part.

When you are over the edge, the cursor feedback will be . This is the feedback for coincident, meaning you are on the edge. Sketch a line to the right.



5 Add a **Tangent Arc**. Draw a tangent arc from the right endpoint of the line to the midpoint of the right vertical edge.



When the cursor is over the right vertical edge, the midpoint will be displayed with a square with two diagonal lines . Once the arc is drawn there will be a midpoint relationship established with the right vertical line.

6 Add a **Tangent Arc** from the left end of the line to the midpoint of the left vertical edge.

7 Dimension the sketch as shown.

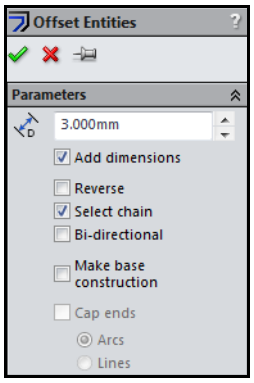


Offset Sketch Entities

The **Offset Entities** tool is used to make a copy of sketch entities, or edges, offset from the original by some specified distance.

8 Click any of the three sketch entities.

9 Click **Offset Entities** on the Sketch toolbar. **Select Chain** will be selected by default, this will select all sketch entities that are continuously connected to the one we selected.



10 Type **3mm** for the offset. Select **Reverse**, if necessary, to make the preview appear above the other sketch entities.

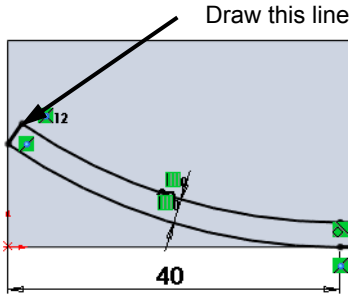


Click **OK**. Each arc and the line are duplicated at an offset distance of 3 mm.

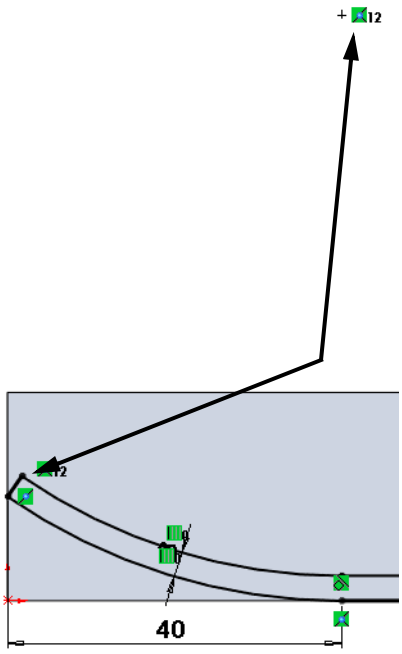
Close the Sketch

The sketch must be closed to extrude the cut. Add two lines to connect the ends of the arcs. These lines need to be perpendicular to the arcs. This means that the lines must point through the centers of their respective arcs.

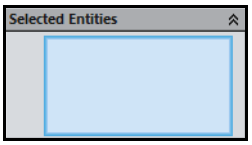
11 Draw a line to connect the ends of the two arcs on the left. The line should become fully defined (black).



12 If the line does not become fully defined, add a relationship to make the line perpendicular to the arc. Because we cannot add a perpendicular relationship between the arc itself and the line we will add a coincident relationship between the line and the centerpoint of the are. By basic geometry this will make the line perpendicular to the arc.



13 Click **Add Relation** on the Sketch toolbar.



14 Whenever the PropertyManager or a dialog box has an entry box colored in the light blue color shown at right, anything selected in the graphics area will be entered in the box.

15 Select the arc centerpoint and the line.

16 Click to add a **Coincident** relationship. The line will now be black as it is fully defined.

17 Repeat the procedure to close the sketch between the arcs on the right.

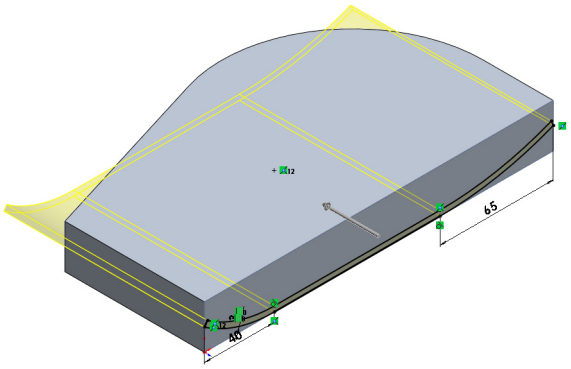
Cut to the Outside

The **Extruded Cut** command can either cut what is enclosed by the sketch or everything outside the sketch.

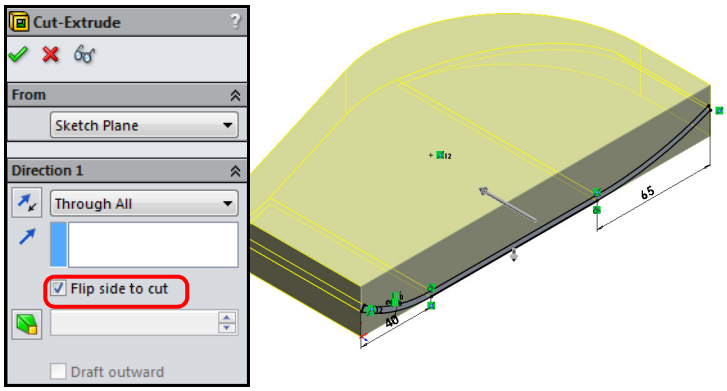
18 Click **Extruded Cut** on the Features toolbar.

19 Select **Through All** for the end condition.

20 By default, the cut will remove the material inside the sketch.

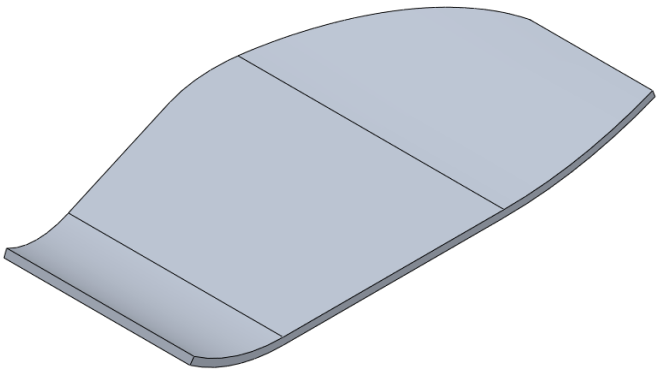


21 Select **Flip side to cut**. Now the cut will keep the material that is inside the sketch and remove the material that is outside the sketch.



22 Click **OK**.

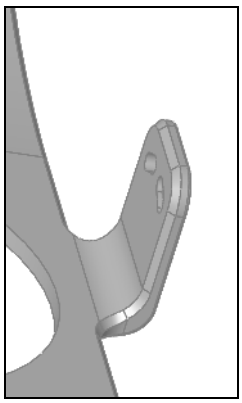
The material remaining is our curved base plate.



23 **Save** the part to the folder `Mountainboard Design Project\Mountainboard\Binding` folder.

Task 3 — Creating the Side Tabs

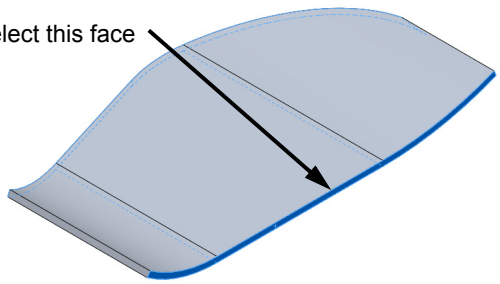
The two side tabs provide the mounting locations for the binding straps that go over the riders foot. They must be offset from the base plate to allow for the thickness of the strap. They must also be a uniform thickness as the final product will be cut from flat material and bent into the final shape.



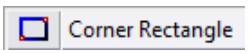
Create First Offset

1 Select the face shown and open a sketch.

2 Change the view orientation to **Normal To** by clicking  on the **Standard Views** toolbar.

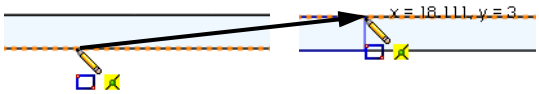


3 Select the **Rectangle** tool  from the **Sketch** toolbar.

Select **Corner Rectangle**  for the **Rectangle Type** in the PropertyManager.

The **Rectangle** tool draws a rectangle with two lines horizontal and two vertical.

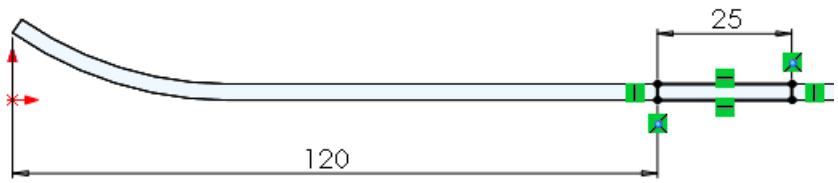
4 Start the rectangle with a coincident relationship to the bottom edge of the `Binding Base Plate`. Drag the rectangle until you get a coincident relationship with the top edge.



5 Add dimensions.

The two coincident relationships control the height and vertical position of the rectangle. To make it fully defined, only a width dimension and a single positioning dimension are need.

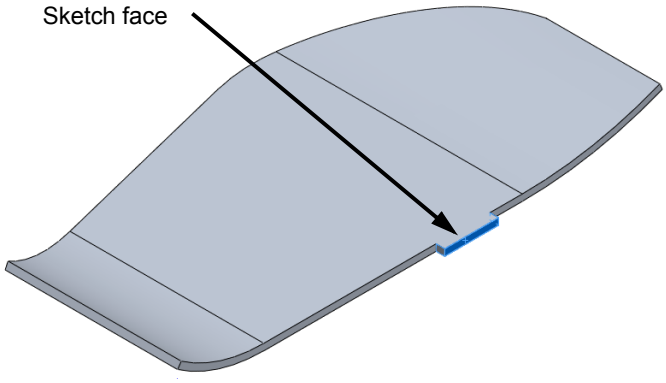
Add the two dimensions shown.



6 **Extrude** the sketch to a blind depth of **4mm**.

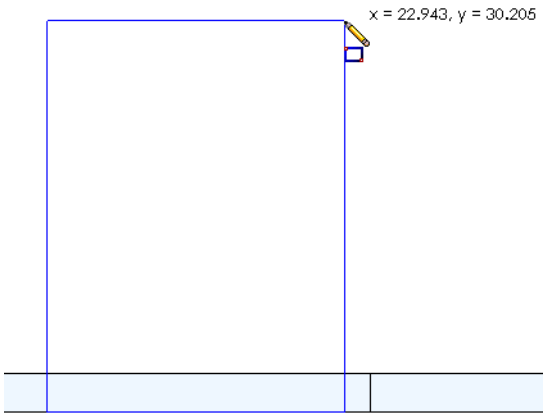
Create the tab

7 Create a new sketch on the new face created by the offset.

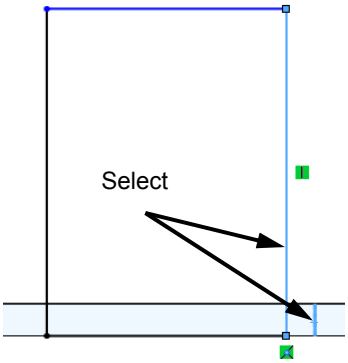


8 Sketch a rectangle. Start the rectangle at the lower left corner of the offset. The cursor feedback will show the yellow coincident callout to indicate we are on the endpoint of the edge.





9 Add a relationship. Click **Add Relation** on the **Sketch** toolbar. Select the right vertical line of the sketch and the right vertical edge of the offset.



10 Click **Collinear** and click **OK**.

11 The rectangle will now stay the same width as the offset.

12 Dimension the height of the rectangle to **32mm**.

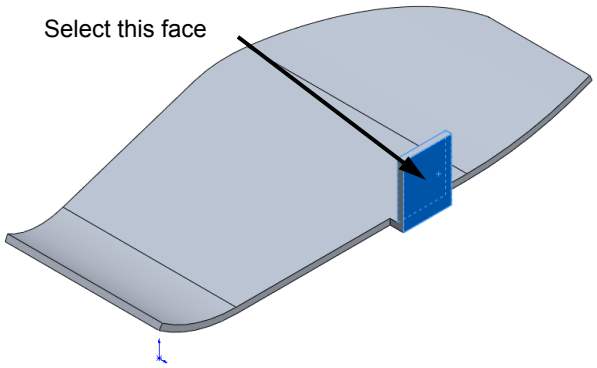
13 Extrude the sketch, **Blind** to a depth of **3mm**.

Task 4 — Binding Attachments

The binding straps will attach to the base plate through two curved slots. The slots need to be curved to allow the binding straps to rotate as the foot is pushed into the binding.

1 Select the outside face of the tab and open a sketch.

2 With the outside face of the tab still selected, click .



Zoom to Selection

To get a closer look at a selected item, click **Zoom to Selection** on the **View** toolbar. This will make the selected entity fill the graphics area.

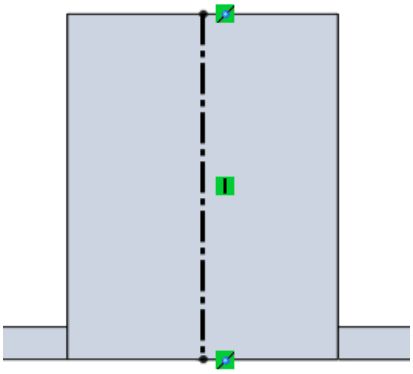
3 Zoom in on the selection by clicking **Zoom to Selection** .

Sketching the Slots

The two slots are symmetrical. To create them, sketch one and mirror it get the second.

4 Create a centerline. Start the centerline at the midpoint of the top edge and make sure it is vertical.

5 To mirror as we sketch, click **Dynamic Mirror Entities** on the Sketch toolbar.



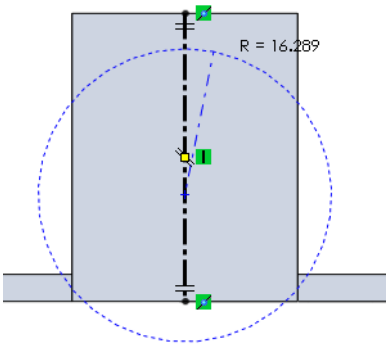
Centerpoint Arc

Drawing the Centerpoint Arc is a two step process. You first drag from the center of the arc to the start point of the arc. Release the mouse button, then press and drag the length of the arc.

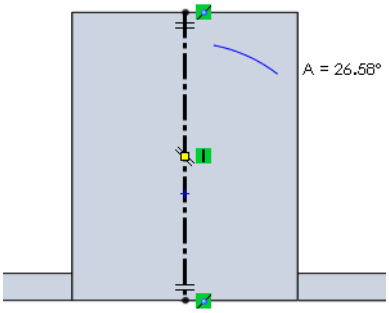
6 Select the **Centerpoint Arc** from the **Sketch** toolbar.

7 Start the arc from the centerline and drag the radius/ start point as shown.

8 Release the mouse button. This will be the point where the arc begins.

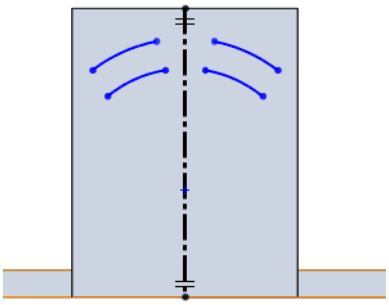


9 Press the mouse button and drag until the arc is as shown, then release the mouse button.



10 Draw another centerpoint arc using the same centerpoint. This makes the arcs concentric.

11 Draw **Tangent Arcs** to close up the sketch. Add Tangent relationships to make sure all the arcs are tangent to the arcs they are connected to.

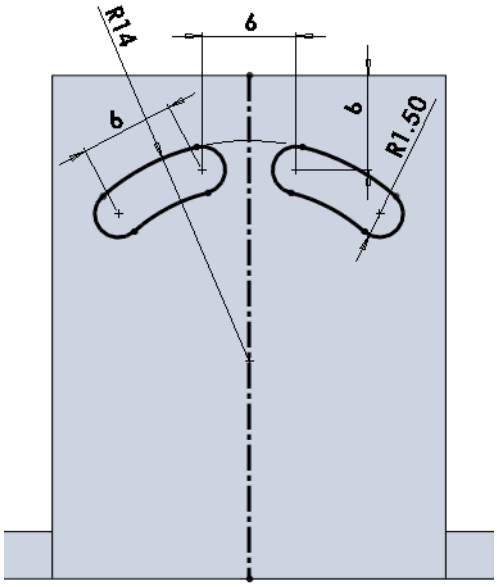


TIP: Toggle on **Sketch Relations** to make it easier to check the relationships.

12 Turn off **Dynamic Mirror Entities**.

Dimension the Sketch

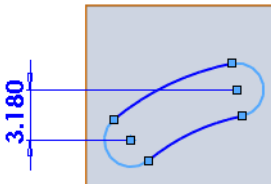
13 Add the Dimensions shown.



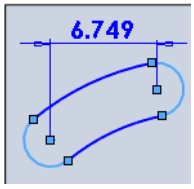
Dimension Alignment

To create the 6mm dimension between the arcs, pick the arcs on each end of the slot. As you drag the cursor, the dimension preview will show:

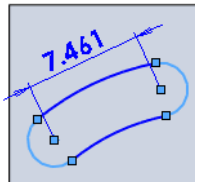
Vertical dimensions



Horizontal dimension



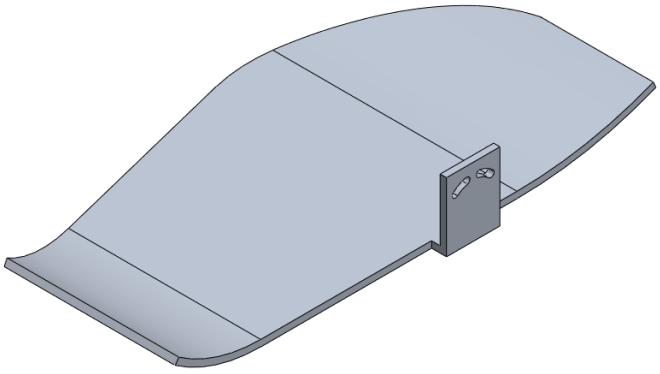
Aligned dimension



To lock in the dimension alignment you want, click the right mouse button to set it. The cursor  shows that clicking the right mouse button will lock the alignment. Once the dimension alignment is locked, the cursor changes to  , to show that you can now unlock the dimension alignment by clicking the right mouse button. Once locked, the alignment of the dimension will stay active no matter where you then move the cursor.

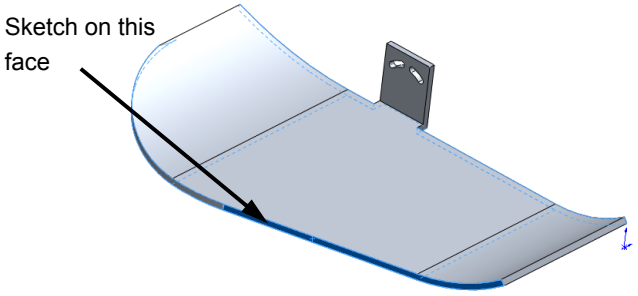
14 Extrude a cut. Select **Up To Next**.

The end condition **Up To Next** will extrude the cut until it reaches the next surface that intercepts the entire profile. In this case it is the inside face of the tab. In some cases, such as this, more than one end condition will produce the same results but for different reasons. We could have used **Through All** and still had the same results.

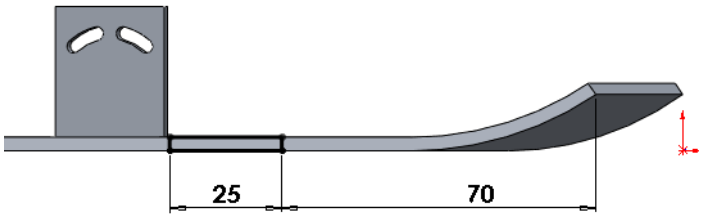


Task 5 — Create the second tab

Create another tab on the other side of the `Binding Base Plate`. Create the offset feature on the face shown.



Position the offset **70mm** from the vertex shown.

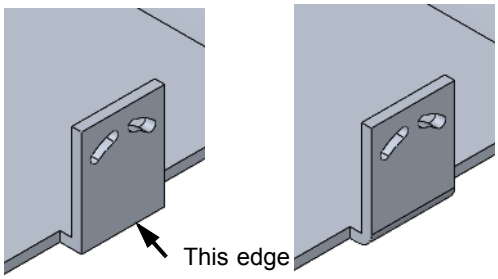


Bends

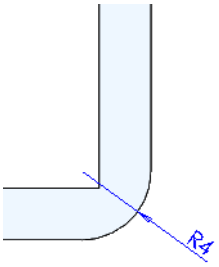
This part will be manufactured from a flat piece of metal. It will first be cut to shape, then the tabs will be bent, followed by the front a back curves. Because the part will be manufactured from a single piece of metal, the model must have a uniform thickness.

We could add the bends manually using fillets.

If we add a **4mm** fillet to the lower edge of the tab.



The profile of the tab shows that the thickness is no longer uniform. We must add a fillet to the inside edge of the tab at the correct radius to keep the material uniform.



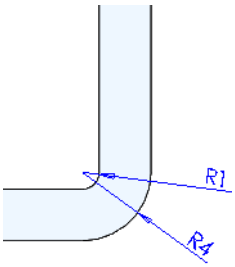
To calculate the fillet radius:

The radius of the inside fillet = radius outside fillet - material thickness.

- Radius $_{inside}$ = Radius $_{outside}$ - Thickness
- Radius $_{inside}$ = 4 - 3 = **1mm**

If we applied a 1mm fillet to the inside edge.

The finished bend would now a uniform thickness.



Sheet Metal

SolidWorks sheet metal functions can create bends from existing square corners and calculate the correct amount of material needed to cut the flat blank.

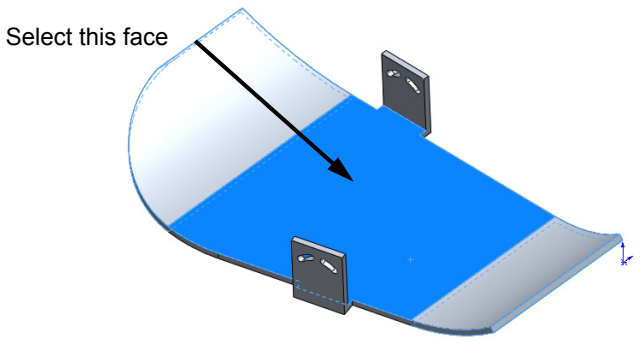
Parts can be flattened to show the correct flat pattern.

Note: Parts must be a uniform thickness in order to insert bends. This is consistent with the process in the shop as the part will be made from a single piece of material that is a consistent thickness.

Task 6 — Add bends

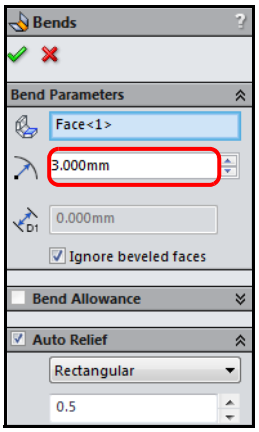
As stated earlier, this part will be created from a single piece of flat metal. Using the Sheet Metal tools, we will add bends to this part so that it can be flattened to determine the size of the blank that will need to be created.

1. Select the top flat face of the `Binding Base Plate`. This will be the face that will be fixed. All the bends will move relative to this face.
2. Click **Insert, Sheet Metal, Bends** from the menu.



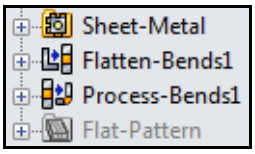
3 Set the bend radius. Change the Bend Radius to **3mm**, which is the material thickness.

Leave the remaining option as shown.



4 Click ✔.

5 Sheet Metal Features. Four new features are added to the FeatureManager design tree:



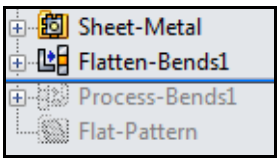
- `Sheet-Metal` contains the sheet metal definitions such as the bend radius we entered in the last step.
- `Flatten-Bends1` creates a flat pattern or the part.
- `Process-Bends1` contains the information to bend the flat pattern into the final part.
- `Flat-Pattern1` also creates a flat pattern of the part. Note that it is gray in color indicating that it is suppressed. This means that the feature is not in use.

Rollback

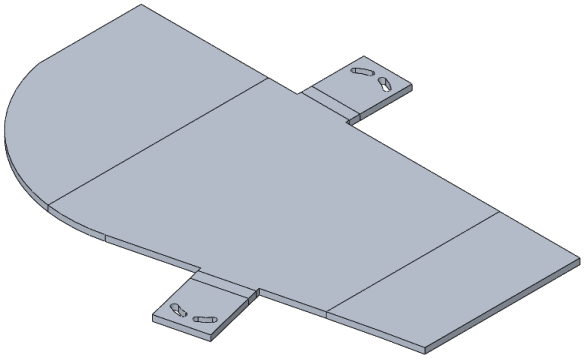
The model can be rolled back to a previous state by moving the rollback bar at the end of the FeatureManager design tree.

To rollback the FeatureManager design tree, move the cursor over the rollback bar (the line that is normally at the bottom of the FeatureManager design tree). The cursor will change to a hand  , then drag the rollback bar to the desired position.

6 In the FeatureManager design tree, move the cursor over the rollback bar and drag the rollback bar to a position between `Flatten-Bends1` and `Process-Bends1`.



The part will flatten.



Working With The Flattened Part

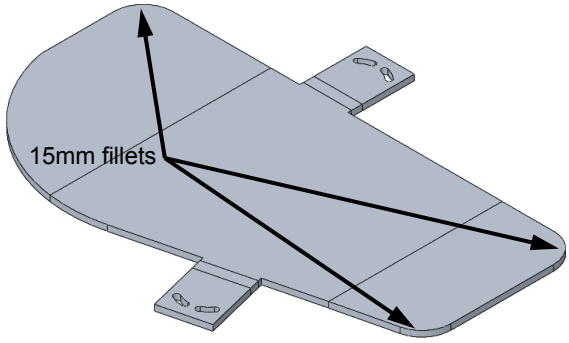
Features can be added to the flattened part just as they can with the bent up part.

Features added in the flattened state usually equate to machining that would be done before the part is bent in the shop.

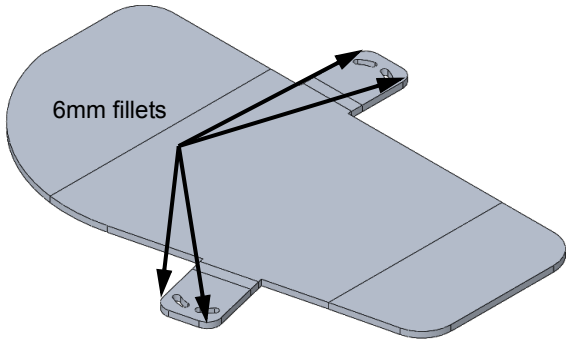
Task 7 — Add Fillets

The sharp corners need to be rounded both to make the part look better and for safety reasons.

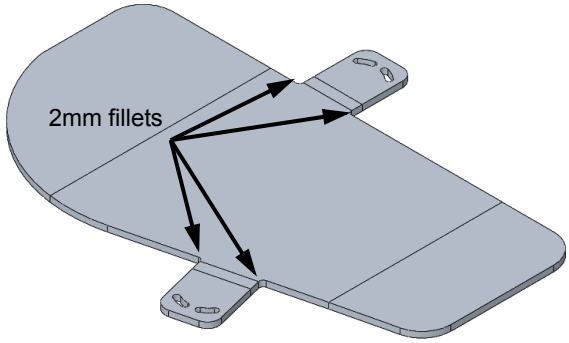
1 Add **15mm** fillets to the corners shown.



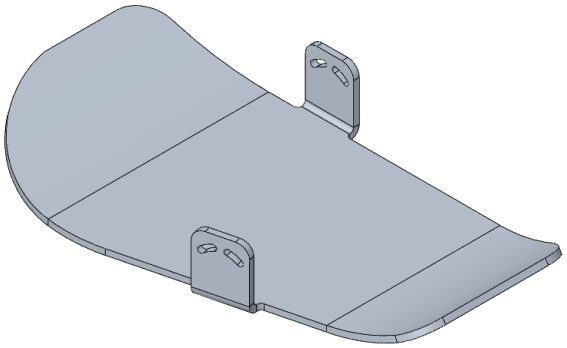
2 Add **6mm** fillets to the corners of the tabs as shown.



3 Add **2mm** fillets to the corners shown.



4 Bend the part by moving the rollback bar to the end of the FeatureManager design tree. The part will bend to its final shape.



Task 8 — Center Hole

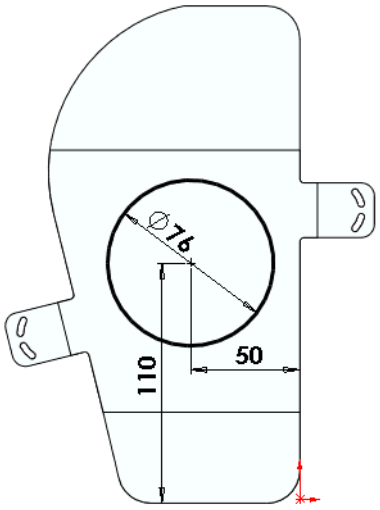
To hold the `Binding Base Plate` to the deck, it will have a hole into which fits the `Binding Anchor` created in the last lesson.

This hole would be created before the part was bent as it is much easier to clamp the part for the drilling operation when the part is flat.

1 Rollback the part to before `Process-Bends1`.

Note: Rollback can also be done through the right mouse button menus. Right-click on `Process-Bends1` and select **Rollback**

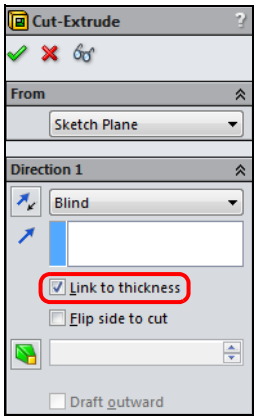
2 Create a sketch on the top face of the part. Sketch a circle and dimension it as shown.



Note: We are cutting the hole just slightly larger than the size of the boss on the `Binding Anchor` part. We want the fit tight, but not so tight that it is hard to assemble.

3 Extrude a cut.

Now that the part has been turned into sheet metal by the **Insert Bends** command, a new end condition appears called **Link To Thickness**. This makes the cut depth the same as the material, even if the material thickness changes.



Note: **Link To Thickness** is only available in parts that have been turned into sheet metal parts by the process of **Insert Bends** or **Base Flange**.

4 Bend the part. Right-click the feature `Process-Bends1` and select **Roll To End**.

This is just another way to move the rollback bar.

